Kaspian Thomas

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EDUCATION

University of New Orleans | New Orleans, LA **Expected May 2027**

B.S. in Computer Science, Concentration in Machine Learning & AI | Cumulative GPA: 3.6 / 4.0 | CSCI GPA: 3.8 / 4.0

- **Relevant Coursework:** Data Structures, Analysis of Algorithms, Natural Language Processing, Machine Structure & Assembly Language, Computer Networks, Database Systems, Python for Data Science & AI
- **Honors:** President's List ×3 (Spring 2025, Fall 2025, Spring 2026); Dean's List (Fall 2024)

PROFESSIONAL EXPERIENCE

HPM Group — Connect AI Initiative / Enterprise Client Integration | New Orleans, Louisiana June 2026-Current

AI Systems Engineer — Agent Platform (Independent Contractor, Part-Time)

Python · REST APIs · Node.js · Linux · systemd · GitHub Actions · SMTP · SSO / zero-trust access · helpdesk and CRM integrations

- Built an automated daily monitoring job that sweeps thousands of IoT devices across multiple customer accounts in about a minute, maps outages to customers and sites, and files a prioritized helpdesk ticket with a color-coded spreadsheet before the workday starts, replacing manual checks.
- Designed an internal operations console (SSO access, server-side token verification, multi-layer role-based permissions, audit logging) and self-supervision (late-run detection, preflight checks, exactly-once ticketing, failure alerts) so support staff can monitor every run without engineering.
- Built a human-approved customer notification pipeline with conservative record matching (zero false matches in validation), on a standard-library-only stack with 1,500+ offline tests, CI, and secret-blocking hooks.

F1 Real-Time Telemetry Commentary System — IBM Hackathon (Co-Author) **May 2026**

Python · FastAPI · WebSocket · React · IBM Granite · Langflow · SQLite · Netlify · *GitHub Repo*

- Co-engineered a gated RAG pipeline distilling ~1,400 lap events per race into ~64 grounded commentary moments (~4.5%), running IBM Granite via Ollama/Langflow over three years of official FastF1 data (64k+ laps) — a 95% reduction in LLM overhead with zero hallucinated race statistics
- Engineered an async WebSocket server in FastAPI broadcasting live telemetry, standings, and commentary to all clients simultaneously; commentary generation runs fire-and-forget so the 1–3s LLM call never stalls the data stream
- Deployed end to end — React dashboard on Netlify, FastAPI backend behind an ngrok static tunnel — with an architecture that migrates from local IBM Granite to IBM watsonx.ai without pipeline changes

PROJECTS

BioBERT Soccer Injury Severity Classifier **Spring 2026**

Python · PyTorch · HuggingFace Transformers · scikit-learn · Matplotlib/Seaborn · Google Colab (T4 GPU) · LaTeX

- Fine-tuned BioBERT to classify soccer injury reports into five severity tiers, training on a Colab T4 GPU
- Built a full evaluation pipeline — ROC/AUC, confusion matrices, precision-recall curves — benchmarked against random and majority-class baselines, and authored the accompanying report in LaTeX

Customer Churn Prediction | CSCI 3220 Final Project **Summer 2026**

Python · Pandas · NumPy · scikit-learn · XGBoost · imbalanced-learn · Jupyter · Git

- Built an end-to-end ML pipeline predicting churn across 7,043 customers, covering EDA, feature engineering, class-imbalance handling, and model selection
- Compared four models under 5-fold stratified cross-validation; handled 26.5% class imbalance with class weighting and scale_pos_weight, applying preprocessing per fold to avoid SMOTE leakage
- Identified the leading churn drivers (month-to-month contracts, fiber service, electronic-check payment) and shipped a reproducible repo with README, report, and evaluation figures

Multiplayer Networked UNO Card Game **Spring 2026**

Java · TCP/IP Sockets · Multi-threading · ExecutorService · Swing

- Implemented a multiplayer UNO server enforcing full rules — special cards, draw logic, turn management — over TCP/IP sockets
- Designed a multi-threaded server on ExecutorService and ConcurrentHashMap with automated matchmaking, game-room management, and countdown timers; built the Swing client with hand rendering, game log, and an integrated chat panel

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, PL/SQL, JavaScript

Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, XGBoost, imbalanced-learn, spaCy, Pandas, NumPy, FastAPI, Pydantic, React, Node.js, Tailwind CSS, Swing

Tools & Platforms: Docker, Git, PostgreSQL, SQLite, Oracle SQL, Langflow, Ollama, Jupyter, Google Colab, Matplotlib, LaTeX, Netlify, ngrok

Methodologies: Agile, Test-Driven Development, CI/CD, Client-Server Architecture, LLM Evaluation & Guardrail Design, Algorithm Analysis